Contents

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Design Document (SDD) is to describe the architecture and system design of the Wordle Game System. This document explains the system components, design structure, database design, and interaction between different parts of the system.

The intended audience of this document includes developers, testers, project managers, and instructors involved in the development and evaluation of the system.

1.2 Scope

The Wordle Game System is an interactive word guessing application that allows users to guess a hidden word within a limited number of attempts. The system provides feedback after each guess to help the user identify correct and incorrect letters.

The main goal of the project is to develop a simple and user-friendly game that improves logical thinking and vocabulary skills while providing an enjoyable user experience.

The objectives of the system include:
- Providing an interactive gameplay experience.
- Validating user guesses and displaying feedback.
- Managing game flow including win and lose conditions.
- Allowing users to restart and play multiple games.

The project benefits include simplicity, ease of use, and providing entertainment for users through a lightweight and interactive application.

1.3 Overview

This document describes the design and structure of the Wordle Game System. It includes the system architecture, data design, component design, interface design, and system diagrams used in the development process.

The document is organized into several sections. The System Overview section provides a general description of the system functionality. The System Architecture section explains the architectural design and system decomposition using diagrams. The Data Design section describes the system entities and database structure. The Component Design section explains the behavior of system components, while the Human Interface Design section describes the user interface and screen interactions.

1.4 Reference Material

- IEEE Std 1016-2009 Software Design Description Standard
- IEEE 830 Software Requirements Specification Standard
- Software Requirements Specification (SRS) Document for the Wordle Game System
- Wordle Game Rules and Gameplay References

1.5 Definitions and Acronyms

Sequence Diagram: A UML diagram that illustrates the interaction flow between system components during gameplay.

ERD (Entity Relationship Diagram): A diagram used to represent entities, attributes, and relationships in the system database.

Player: The user who interacts with the system and plays the game.

Authentication: The process of verifying the player’s login information before accessing the game.

List of Words: A collection of predefined valid words used by the system during gameplay.

Word: The hidden word that the player attempts to guess.

Game: A game session in which the player tries to guess the hidden word within a limited number of attempts.

Attempt: A single guess entered by the player during the game session.

2. SYSTEM OVERVIEW

The Wordle Game System is an interactive word guessing application designed to provide users with an enjoyable and simple gameplay experience. The system allows players to log into the application, start a new game session, and attempt to guess a hidden word within a limited number of attempts.

The system validates user guesses and provides feedback after each attempt, indicating whether letters are correct, misplaced, or incorrect. The game continues until the player correctly guesses the hidden word or reaches the maximum number of allowed attempts.

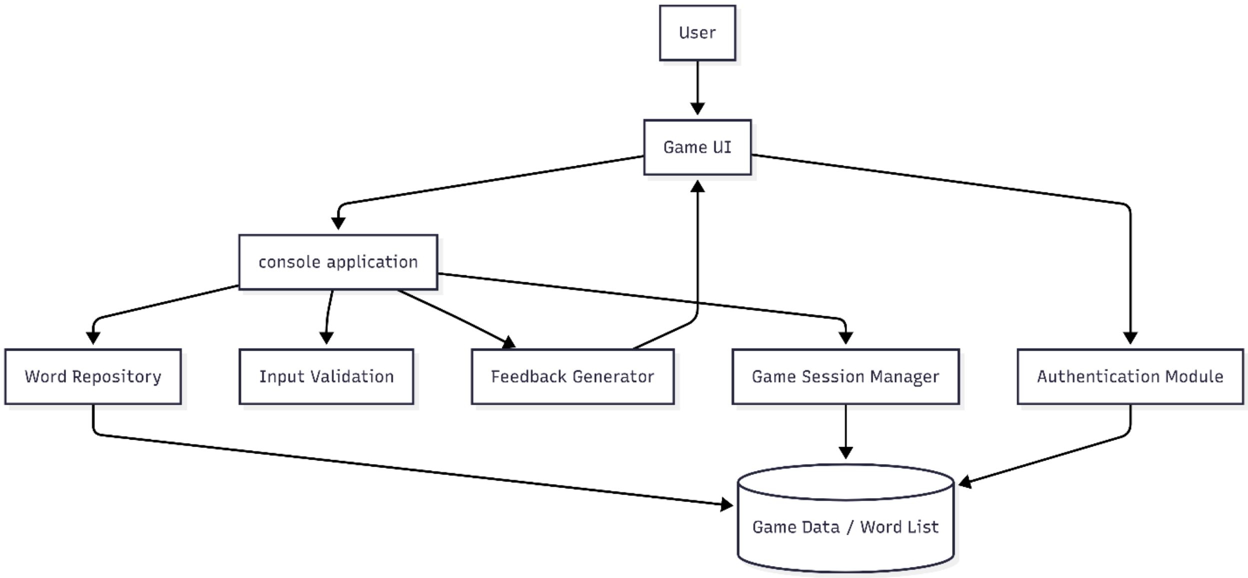
The project follows a modular design consisting of several main components, including the Game Interface, Authentication module, Game Controller, and Word List module. These components work together to manage gameplay, validate input, and display results to the player.

The system is intended to be lightweight, user-friendly, and easy to maintain. It is designed for users with basic computer skills and does not require external systems or internet connectivity during gameplay.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

- **Architecture Diagram**



3.2 Decomposition Description

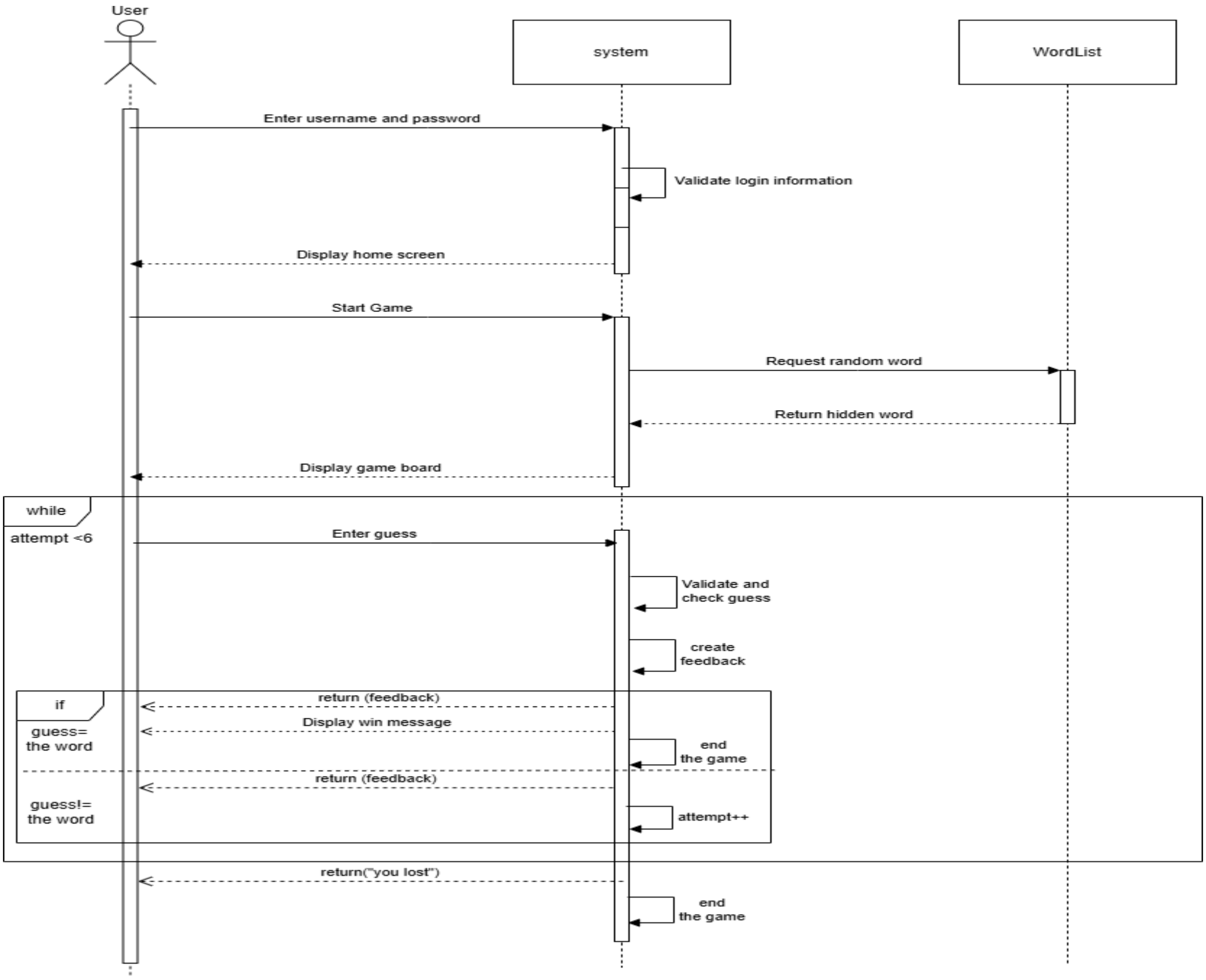
The system is decomposed into several main components that work together to provide the required functionality of the Wordle Game System.

The following diagrams are used to describe the system decomposition and interaction between components:

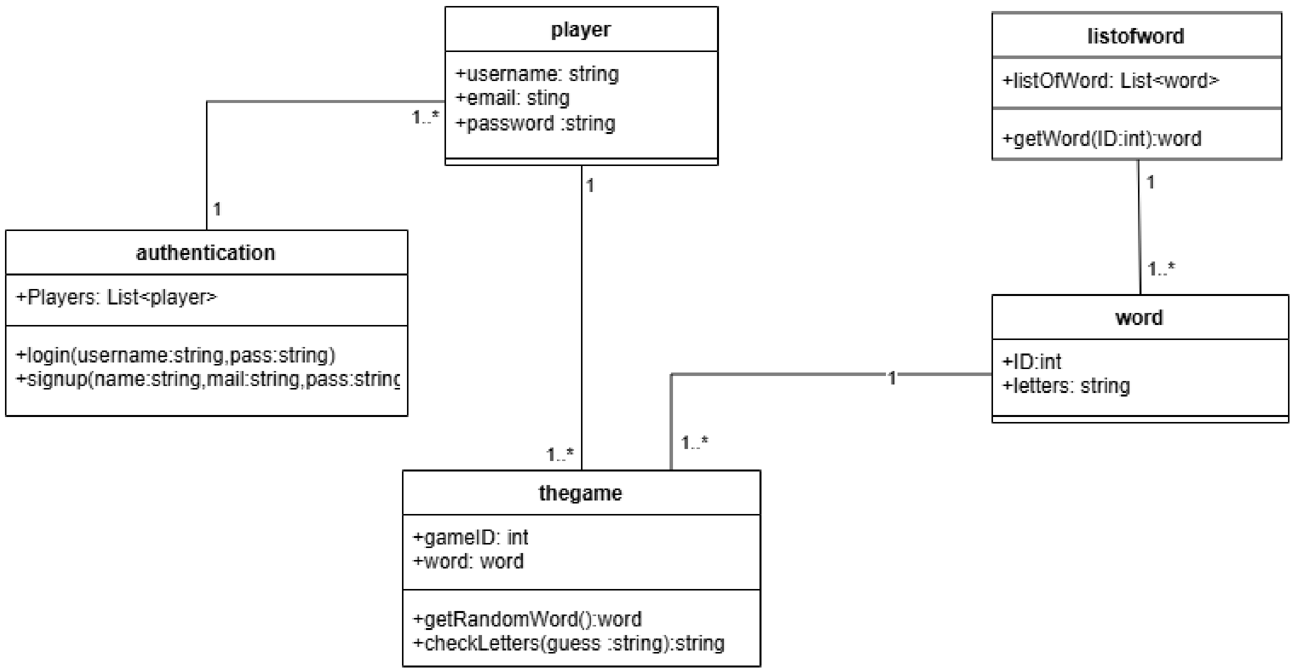
- Sequence Diagram: Illustrates the sequence of interactions between the player, game interface, authentication module, game controller, and word list during gameplay.

- Class Diagram: Represents the main classes in the system, including Player, Game, Attempt, Word, Authentication, and Word List, along with their attributes, methods, and relationships.

- **Sequence Diagram**


User
system
WordList
Enter username and password
Validate login information
Display home screen
Start Game
Request random word
Return hidden word
Display game board
while
attempt <6
Enter guess
Validate and
check guess
create
feedback
if
return (feedback)
guess=
the word
Display win message
end
the game
return (feedback)
guess!=
the word
attempt++
return("you lost")
end
the game

Class Diagram



4. DATA DESIGN

4.1 Data Description

The data design of the Wordle Game System describes how player information, game sessions, words, and gameplay attempts are organized and managed within the system.

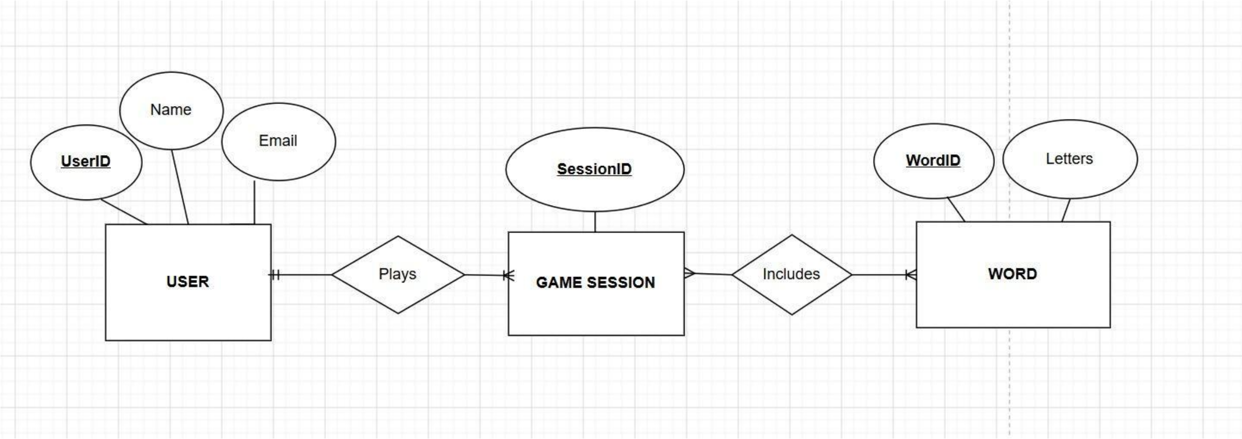
The system consists of the following main entities:

- User: Stores player information such as user ID, name, and email.
- Game Session: Represents a single gameplay session played by the user.
- Word: Stores the hidden words used during gameplay.
- Attempt: Represents each guess entered by the player and the corresponding feedback.

The relationships between the entities are as follows:

- One user can play multiple game sessions.
- Each game session uses one hidden word.
- Each game session contains multiple attempts.
- Each attempt belongs to one game session.

The ER Diagram below illustrates the entities, attributes, and relationships used in the system database design.


Name
Email
UserID
SessionID
WordID
Letters
USER
Plays
GAME SESSION
Includes
WORD